

Navigator YouTube MCP - API Client Documentation

1. Overview

Navigator YouTube MCP is a private, internal-use MCP (Model Context Protocol) server that connects to the YouTube Data API v3 via OAuth 2.0. It is used exclusively by the channel owner of the Navigator Podcast YouTube channel (<https://www.youtube.com/@navigatorpodcast>) to manage channel content programmatically. The tool is not publicly accessible and has a single authenticated user.

2. Architecture

The MCP server acts as a bridge between the channel owner's workflow tools and the YouTube Data API v3. Authentication is handled via OAuth 2.0 with the channel owner's Google credentials. The server runs locally and makes API calls on behalf of the authenticated user to their own channel only.

Component	Description
MCP Server	Local server running pauling-ai/youtube-mcp-server
Authentication	OAuth 2.0 (Google credentials, Brand Account)
Google Cloud Project	Navigator YouTube MCP (ID: api-project-759297478740)
YouTube Channel	Navigator Podcast (@navigatorpodcast)
User Count	1 (channel owner only)

3. YouTube API Endpoints Used

API Endpoint	Purpose	Est. Daily Calls
youtube.videos.list	List channel videos, verify episode numbering	2-5
youtube.videos.update	Update titles, descriptions, tags for SEO	2-10
youtube.playlists.list	List existing playlists	1-2
youtube.playlistItems.insert	Add videos to playlists	1-5
youtube.playlistItems.delete	Remove videos from playlists	0-2
youtube.commentThreads.list	Read viewer comments for engagement	1-3
youtube.captions.list	List available caption tracks	1-5
youtube.search.list	Search channel content	0-2

4. Primary Use Cases

- Episode Metadata Management:** The channel has 46+ full podcast episodes and 150+ short clips. The MCP server is used to list all videos and update titles, descriptions, and tags to maintain consistent formatting (e.g., episode numbering pattern: EP01, EP02, etc.).
- Playlist Organization:** Episodes are organized into series-based playlists (e.g., '7 Habits' series, 'KAW/Betenbough' series, 'Community' series). The API is used to add and remove videos from playlists.
- Comment Monitoring:** The MCP server retrieves viewer comments for engagement analysis and community interaction.
- Transcript Retrieval:** Caption tracks are accessed for content analysis, episode synthesis, and documentation.
- SEO Optimization:** Video metadata is updated based on channel performance analysis to improve discoverability.

5. Quota Usage Estimate

The channel owner uses the MCP server 1-2 times per week in focused sessions. Estimated peak daily usage is under 2,000 quota units. The requested 10,000 units/day provides comfortable headroom for batch operations (e.g., updating multiple video titles at once).

Operation	Quota Cost	Frequency	Daily Units
videos.list (50 results)	~100	2-5x	200-500
videos.update	~50 each	2-10x	100-500
playlists.list	~1-2	1-2x	2-4
playlistItems.insert/delete	~50 each	1-5x	50-250
commentThreads.list	~1-2	1-3x	2-6
captions.list	~50	1-5x	50-250
		TOTAL (peak)	~1,500

6. Data Handling & Compliance

All YouTube API data retrieved by the MCP server is used transiently within the current session and is not persisted beyond 24 hours. The API Client does not commercialize YouTube data, does not display data across multiple platforms, and does not create reports from YouTube data for external distribution. All operations are strictly for internal channel management by the single authenticated channel owner.